Operational Memo -- Scenario C: Governance Across 14, 6, and 0 Crew Regimes

To: SDV Program Office & Vessel Operations Command

From: Principal Systems Architect, sdv-resource-governor

Subject: Deterministic Resource, Deck-Safety, and Autonomy Rules Under Crew Regimes 14, 6, and 0

1. Executive Summary

A Software-Defined Vessel (SDV) cannot treat human crew availability as a mere scheduling annotation: crew count directly governs which physical payload transitions are admissible, which manual deck evolutions can be staffed, whether at-sea vehicle replacement (A' window) is physically safe, and how fault recovery escalates when telemetry degrades or a payload overconsumes. In sdv-resource-governor, human crew availability, flight-deck clearance (deck_clear), and port-only installation constraints are enforced as hard operational guards inside scheduling.py, admission.py, and replay.py.

2. Quantitative Regime Comparison (FULL_CREW_14 vs. REDUCED_CREW_6 vs. UNCREWED_0)

Operational Dimension	Regime 1: 14 Crew (FULL_CREW_14)	Regime 2: 6 Crew (REDUCED_CREW_6)	Regime 3: 0 Crew (UNCREWED_0)
Bridge & Engine Watch	5 personnel	4 personnel	0 (Remote Shore Control + Onboard Autonomy)
Available Deck/Payload Crew	9 personnel	2 personnel	0 personnel
Max Concurrent Manual Deck Teams	2 teams (for non-conflicting manual tasks)	1 team (strictly serialized)	0 manual deck evolutions
G4 UAV Launch / Recovery (5-12 kn into wind, wind <= 25 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; 15-60 s local bridge confirmation)	PERMITTED (2 deck crew; 60-180 s watch handover confirmation)	FORBIDDEN (requires manual flight-deck handling; only stowed/standby modes)
G2 Winch / Probe Deployment (0-5 kn, wind <= 20 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; cannot overlap G4 flight ops)	PERMITTED (2 deck crew; serialized)	FORBIDDEN (requires umbilical/winch deck handling)
Simultaneous G2 Winch + G4 Flight Ops	FORBIDDEN (Annex A.5 requires flight deck clear during G4 ops; speed windows 0-5 kn vs 5-12 kn only touch at the 5.0 kn boundary and require conflicting station-keeping vs into-wind heading)	FORBIDDEN (insufficient crew + Annex A.5 deck clearance + conflicting heading/station-keeping profiles)	FORBIDDEN (0 crew)
At-Sea Vehicle Replacement A' (4-8 kn, wind <= 20 kn, SS <= 3; G4: 3 crew/30m, G3: 4 crew/45m)	PERMITTED (available_crew=9 >= 4 for G3 and >= 3 for G4)	BLOCKED (available_crew=2 < 3 for G4 and < 4 for G3 safe replacement at sea)	FORBIDDEN (0 crew; all vehicle replacements deferred to port)
At-Sea Bay/Mast or >250 kg Module Install (G5, G6 420 kg in bay2/bay1_a)	FORBIDDEN AT SEA (Annex A.5 bay/mast installs are port-only; Annex A.4 limits at-sea packages to <= 250 kg)	FORBIDDEN AT SEA (port-only bay/mast + >250 kg + available_crew=2 < 4)	FORBIDDEN AT SEA (CRW-02: zero physical installations at sea)
Autonomous Sensor & Compute Modes (G1, G3 already in-water or in-cradle charge, G5, G6 port-installed)	100% Operational	100% Operational	Operational for software-governed modes (mode step-down, throttle, power shed); physical stern-ramp launch/recovery of G3 requires 4 deck crew and is FORBIDDEN under 0 crew

3. Concrete Operational Rules & Physical Interlocks Enforced by the Solver

* 1. Why G2 and G4 Cannot Execute Simultaneous Deck Operations Even With 14 Crew (Annex A.5):
  * While 14 crew provides 9 deck/payload personnel (enough to staff two manual handling teams for non-conflicting tasks), G2 ROV/winch deployment and G4 UAV launch/recovery are physically incompatible: (a) Annex A.5 mandates that "Launch or recovery of a drone requires the flight deck to be clear of other ongoing deck operations" (state.deck_clear == True), and (b) G2 requires low-speed station-keeping (0-5 kn) whereas G4 requires 5-12 kn with vessel heading aligned within +/- 15 deg of the wind (their speed ranges only touch at the single 5.0 kn boundary, where deck_clear and heading/winch-handling rules still strictly forbid overlap).
* 2. Why G6 (420 kg, bay2/bay1_a) Is Strictly Port-Only Across All Crew Regimes (Annex A.4 & A.5):
  * Annex A.5 specifies that payload installation in Bays 1-2 and Mast Positions 1-3 is a port-only operation, and Annex A.4 restricts at-sea A' package installation to light deck packages <= 250 kg in Sea State <= 3, wind <= 20 kn, and 4-8 kn. Because G6 weighs 420 kg > 250 kg and installs into bay2 / bay1_a, G6 has port_only: true and is rejected (NO / NOT_NOW) for at-sea installation under 14, 6, and 0 crew alike.
* 3. Acoustic Survey Deck-Machinery Lockout (G3 forbids_deck_machinery_while_active=True):
  * Across both 14 and 6 crew regimes, while G3 is in SURVEY_QUIET (1 kW, 2-6 kn), hydraulic A-frames, winches (G2), and flight-deck handling gear (G4) are locked out to prevent acoustic corruption of the seabed survey sonar.
  * Before turning into the wind or operating deck machinery for G4 recovery, the governor must issue ACOUSTIC_HOLD to G3, verify a causal ACOUSTIC_HOLD_CONFIRMED ACK from guest_id='G3' matching the active command_id and plan_version (15 s nominal acoustic handshake, 60 s timeout), and confirm G3 has transitioned to HOLD_ACOUSTIC.
* 4. Separation of Normal Flight Window vs. A' At-Sea Replacement Window (CRW-03):
  * Normal G4 Flight Launch/Recovery: vessel speed 5-12 kn, wind <= 25 kn, Sea State <= 4, heading aligned within +/- 15 deg of wind, deck_clear=True, 2 deck crew (permitted under 14 and 6 crew).
  * A' At-Sea Vehicle Replacement: vessel speed 4-8 kn, wind <= 20 kn, Sea State <= 3, deck_clear=True, 3 deck crew / 30 min for G4 or 4 deck crew / 45 min for G3 (permitted only under 14 crew; blocked under 6 and 0 crew).

4. Fault Governance & Decision Escalation Under Reduced (6) and Uncrewed (0) Regimes

When operating with 6 or 0 crew, manual intervention on tripped breakers, overconsuming payloads, or degraded cooling loops is delayed (6 crew: 60-180 s watch handover) or impossible (0 crew). The governor enforces three automated fail-safe barriers:

* Urgent Safety Commands (STOP, HOLD, SECURE <= 200 ms, SAF-01): Routed strictly over the dual 10 Gbps local shipboard LAN (< 5 ms transport latency), never over 600-800 ms SATCOM. Measured execution latency on both MacBook Pro (arm64) and the live Arduino Ventuno Q (training06, aarch64) is < 1 ms (<< 200 ms deadline).
* Guest Overconsumption (+10% to +20%, RUN-04): Because none of G1-G5 is safety-critical by design, if a payload (e.g., G5 declared 45 kW drawing 54 kW = +20%) exceeds its manifest envelope, 1:1 thermal coupling immediately adds +9 kW to the chilled-water ledger and the governor autonomously throttles G5 to DEGRADED_70PCT_LOSS (8 kW) while protecting active uninterruptible/recovery tasks (G3 recharge, G4 flight/recovery).
* Lost or Stale Telemetry Feeds (RUN-03): If seawater_temp_c exceeds its TTL (STALE / UNAVAILABLE), the governor never extrapolates optimistic cooling; it clamps to the conservative 30 deg C (120 kW nominal / 72 kW degraded) thermal bound until sensor freshness is restored.